

Transfer Learning Principles

Why representations learned on Task A accelerate learning on Task B — the foundation of all fine-tuning

9 min

Duration

B3 – Apply

Bloom's

L2 – Intermediate

Level

AI-FT-IN-TH-000005

ID

Transfer Learning — The Core Hypothesis

Feature-Reuse Hypothesis

A model trained on Task A learns internal representations that are useful for Task B, reducing the data and compute needed for Task B. The more similar Tasks A and B are, the more effective the transfer.

- Transfer learning is why fine-tuning with 10K examples can match training from scratch with 10M examples
- The pre-trained model is a compressed representation of vast world knowledge and language structure
- Fine-tuning re-purposes these representations for a specific distribution — the new task 'rides on' prior learning
- Analogy: A medical school graduate (domain expert) learns a new specialty faster than a layperson learns medicine

What Each Transformer Layer Learns — Feature Hierarchy

Layers 1–3

Tokenisation artefacts, character patterns, punctuation, basic morphology. Highly general — transfer to almost any task.

Layers 4–8

Part-of-speech, syntactic dependencies, co-reference. Still broadly transferable across most language tasks.

Layers 9–16

Entity types, word sense disambiguation, semantic roles. Useful for many NLP tasks but starting to specialise.

Layers 17–24

Domain facts, reasoning patterns, task-specific representations. Most change occurs here during fine-tuning.

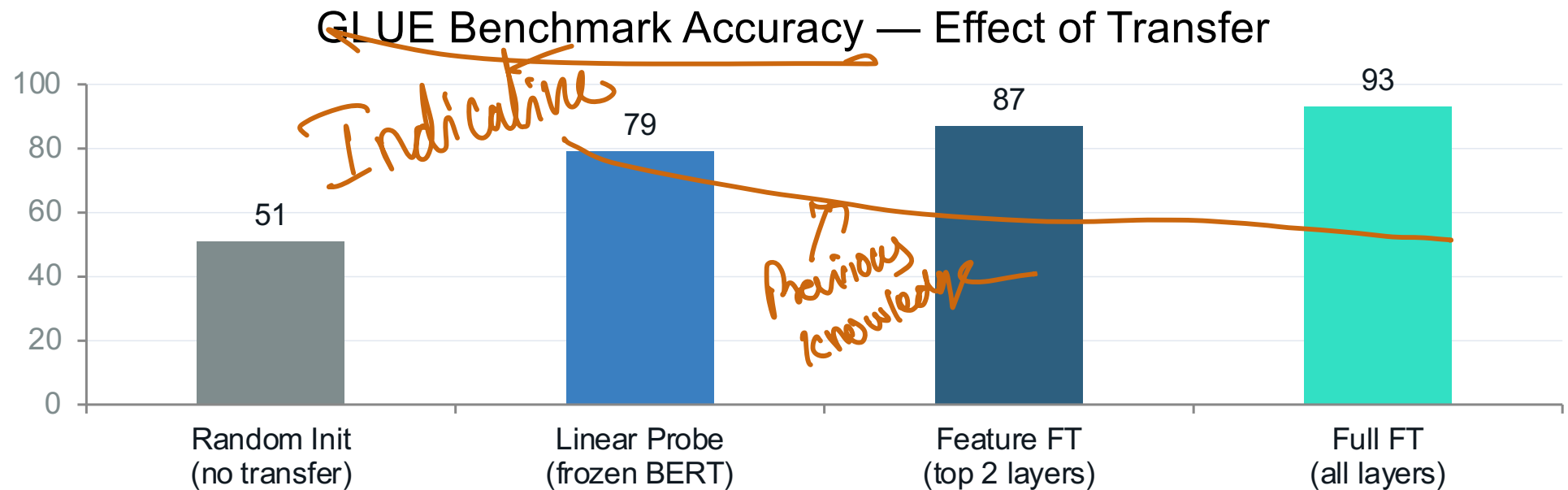
Final Layer

Task-specific predictions (logits). Always retrained. LM head replaced for classification tasks.

Linear Probing Experiment — Measuring Transfer

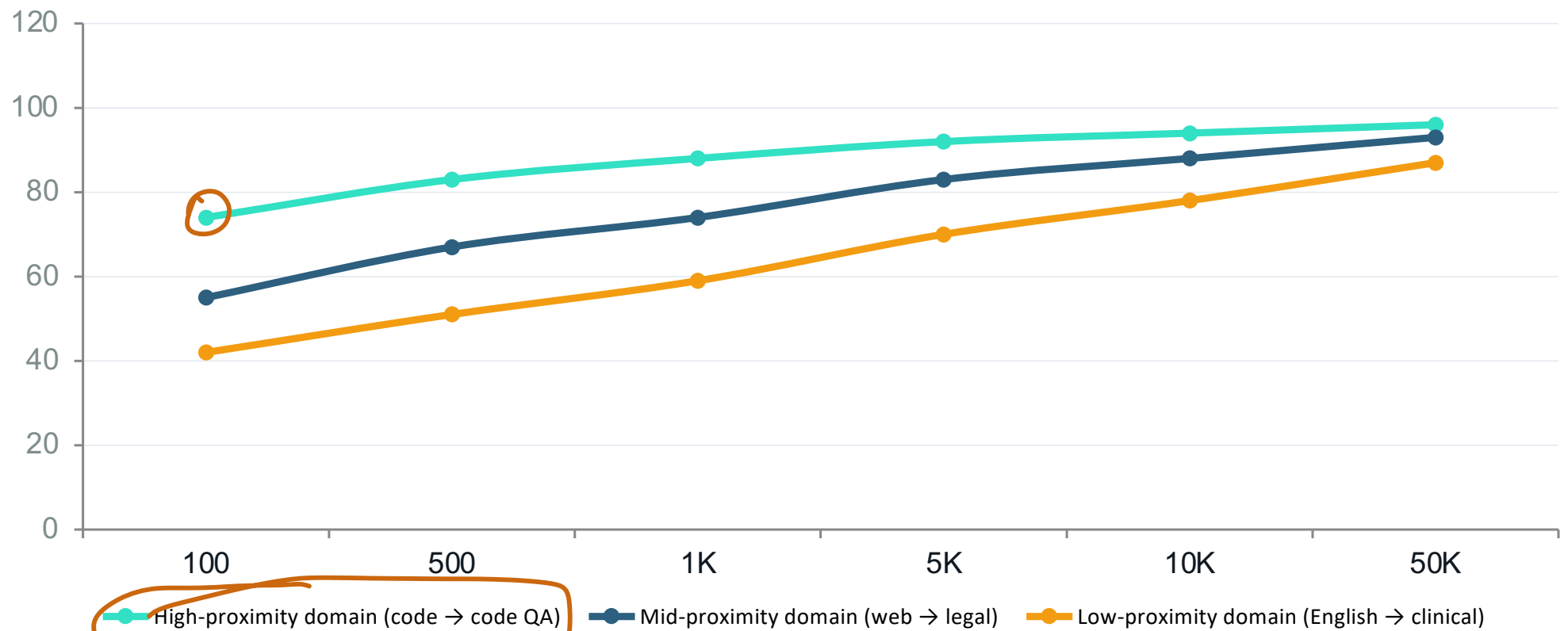
Linear Probing Setup

Freeze the pre-trained encoder completely. Train only a single linear classification head on top. Performance measures how much task-relevant information is already in the frozen representations — no fine-tuning of the backbone at all.



Domain Proximity & Transfer Effectiveness

Fine-Tune Performance vs Training Examples (by domain proximity)



Zero-Shot vs Few-Shot vs Fine-Tuned Regimes

Regime	Training Examples	Mechanism	Typical Performance	Use When
Zero-shot	0	Pure transfer from pre-training	60–80% of fine-tuned	No labelled data available
Few-shot	5–50 in prompt	In-context learning, no weight update	70–85% of fine-tuned	Fast iteration, small datasets
Fine-tuned	500–100K	Weight updates on task data	Baseline (100%)	Production, consistent quality
Full fine-tune	100K+	All weights updated	100–103% of LoRA FT	Large budget, max performance

Practical Model Selection — Domain Proximity Guide

Base model close to target domain

Use a domain-adjacent base (CodeLlama for code, BioMedLM for bio).
Need 10× fewer examples to reach target performance.

Base model distant from target

Start with a general instruction-tuned model (Llama 3 Instruct, Mistral Instruct). Need more data but benefits from RLHF alignment.

When to use linear probing first

Run linear probing before full fine-tuning to validate that the base model's representations are transferable. Saves compute on dead-end experiments.

Few-shot as upper bound estimator

Run few-shot prompting first to estimate what fine-tuning performance ceiling might be. If few-shot is already 90%, fine-tuning likely adds modest gains.

KEY REFERENCES

- Yosinski, J. et al. (2014). How transferable are features in deep neural networks? NeurIPS 2014.
- Devlin, J. et al. (2018). BERT: Pre-training of Deep Bidirectional Transformers. NAACL 2019.
- Raffel, C. et al. (2020). Exploring the Limits of Transfer Learning with T5. JMLR 2020.

 **Next: Concept 6: Full Fine-Tuning Implementation (AI-FT-IN-CD-000001)**